

LAB TASK-11

Assignment: Difference Between Key AI Concepts

Introduction

Modern Artificial Intelligence systems use different tools, models, and techniques to process data, retrieve information, and generate outputs. This assignment explains and differentiates important concepts used in Generative AI and machine learning systems.

1. LangChain

LangChain is a development framework used to build applications powered by Large Language Models (LLMs). It connects LLMs with external tools, memory, APIs, and databases. It is mainly used to build intelligent applications like chatbots, AI agents, and RAG systems.

2. LLMs (Large Language Models)

LLMs are deep learning models trained on massive text datasets to understand and generate human-like language. They can perform tasks such as answering questions, writing content, summarizing text, and translating languages. Examples include GPT models.

3. RAG (Retrieval-Augmented Generation)

RAG is an AI technique that improves LLM performance by combining retrieval and generation. First, it retrieves relevant information from external sources, then uses an LLM to generate a more accurate and context-aware response.

4. FAISS (Facebook AI Similarity Search)

FAISS is a library developed by Facebook for efficient similarity search in large datasets of vectors (embeddings). It is widely used in semantic search and RAG systems to quickly find similar documents.

5. Vector

A vector is a numerical representation of data (text, image, or audio) in multi-dimensional space. In AI, vectors are used to measure similarity between different pieces of data.

6. Vector Database (VectorDB)

A Vector Database is a specialized database designed to store and search vector embeddings efficiently. It allows fast similarity searches in AI applications. Examples include Pinecone, Weaviate, and ChromaDB.

7. Generative AI (GenAI)

Generative AI refers to AI systems that can create new content such as text, images, audio, or code. These systems learn patterns from training data and generate original outputs. Examples include ChatGPT and DALL·E.

8. GANs (Generative Adversarial Networks)

GANs are deep learning models consisting of two networks:

- **Generator:** Creates fake data
 - **Discriminator:** Detects whether data is real or fake
- Both networks compete, improving each other and producing highly realistic outputs, especially in image generation.

Conclusion

These concepts are foundational in modern AI systems. LLMs generate language, RAG improves accuracy using external data, FAISS and VectorDB manage similarity search, and GANs generate realistic synthetic data. Together, they form the backbone of advanced AI applications.